

signal we use the number 1 from the binary code in the 16th pin of the GPIO. This is stored in the registry r1 at the address 40 to the base 10 added to the GPIO controller address. This location happens to be the address which would turn the pin off and activate the light.

Signing Off Code

Although we have successfully activated the light on the LED the processor still needs to be told the final steps. The processor is designed to be active as long as it is powered and needs to be given instructions on what to do next otherwise it will crash. In order to prevent this we need to give it an infinite task which is accomplished by the following code:

loop\$:

b loop\$

This code isn't a command but a label which names the next line "loop\$" giving the line a name. Labels are used only for code reading purposes by humans and are not converted into binary. As per protocol, the dollar symbol is used to denote labels which are only relevant to the local block of code as opposed to the whole file. We have therefore labeled the first line of code as "loop". The "b" command in the second line or the branch command causes the line of code to be executed to be the one that has the "loop" label instead of any line written after it. This causes the processor to go back to the first line which was named "loop" and the cycle goes on forever in the processor. Keep in mind that the last line in the assembler code should be a blank line since the GNU Toolchain tool expects assembly code files to end in a blank line. Otherwise the software assumes the code isn't finished and it was an accidental gap prompting you for action.

Loading the OS

Believe it or not, if you have followed all the instruction so far, you have successfully written your first operating system. Now all that remains is to load it on to the Raspberry Pi. From within GNU Toolkit go to the parent directory of the source directory where the code is written. Type "make" and press enter - this should generate three files - kernel.img, kernel.list and kernel.map. The first is the compiled image of your operating system, with the second being the listing of the assembler code and the final one being the map of where all the labels are placed.

To install this image file of your operating system you need to take the SD card for your Raspberry Pi. It is assumed that the SD card in use